



Cover illustration

Special issue *Biopharmaceuticals*

Drug discovery and production are highly interlinked, because the main product attributes need to be defined early on. This Special issue edited by Alois Jungbauer (University of Natural Resources and Life Sciences, Vienna, Austria) and Klaus Graumann (Sandoz GmbH, Kundl, Austria) covers several areas relevant to the discovery, development and manufacturing of biopharmaceuticals. Topics include antibodies, protein engineering and purification. Images: Antibodies © Elena Pankova – Fotolia.com; Biopharmaceutical Production © Sandoz.

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Research Article

Periodic counter-current chromatography – design and operational considerations for integrated and continuous purification of proteins

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Integrated and continuous processing of recombinant proteins offers several advantages over batch or semi-batch processing used traditionally in the biotechnology industry. This paper presents a theoretical and practical approach for designing a periodic counter-current chromatography (PCC) operation as a continuous capture purification step that is integrated with a perfusion cell culture process. The constraints for continuous and optimal PCC operation govern the selection of residence time and number of columns. The flexibility available in PCC design for selection of these parameters is dictated by the binding characteristics of the target protein on the capture resin. Using an empirical model for the protein breakthrough curve, analytical solutions to determine these conditions were derived and verified with experimental results for three different proteins: two relatively unstable proteins (recombinant enzymes) and a relatively stable protein (monoclonal antibody). The advantages of a continuous downstream capture step are highlighted for the three case studies in comparison with the existing batch chromatography processes. The use of PCC leads to improvements in process economics due to higher resin capacity utilization and correspondingly lower buffer consumption. Furthermore, integrated and continuous bioprocessing results in a smaller facility footprint by elimination of harvest hold vessels and clarification, as well as by reducing the capture column size by one to two orders of magnitude.

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1 Introduction

The steel, petrochemical, chemical, food and pharmaceutical industries have practiced continuous manufacturing for a number of years [1–6]. Biotechnology is late to embrace continuous processing mainly due to the fact that innovation in the industry has traditionally been more product than process centered. To date, studies of con-

tinuous processing of recombinant proteins have mainly focused on either upstream [7–10] or downstream [11–13], but not on both in an integrated fashion. Primarily, this is because the industry is currently dominated by monoclonal antibody (mAb) manufacturing in batch or fed-batch bioreactors [14–16]. Continuous downstream capture studies have demonstrated the significant economic advantages that result from reduced buffer usage, higher capacity utilization and decreased column volume. However, the true potential of continuous chromatography can only be realized by integration with a continuous upstream process [17–19].

In the last few years, simulated moving bed (SMB) systems have been designed by Novasep (Pompey, France),

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Abbreviations: PCC, periodic counter-current chromatography; HIC, hydrophobic interaction chromatography; mAb, monoclonal antibody; R-R, recover-regen (zone)

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Tarpon (Worcester, MA, USA), Semba (Madison, WI, USA), Contichrom (Knauer, Germany) and GE Healthcare (Piscataway, NJ, USA). These systems are capable of the continuous capture and purification of biologics such as recombinant proteins. In this paper, we describe the theory, design and use of periodic counter-current chromatography (PCC) for recombinant protein purification. The PCC system was designed by GE Healthcare based on a customization of the ÄKTATM platform and the creation of a novel control strategy based on UV measurements [11, 20]. In PCC, the target protein breakthrough from one column is captured on a second column, resulting in much higher resin capacity utilization than batch chromatography [21]. Here we expand our previous work on creating a novel biomanufacturing platform [22] by describing the design and operation of the PCC capture of three target proteins – two recombinant enzymes and an mAb. The term batch chromatography is used in this paper to refer to batch-wise single column chromatography.

To understand the design and operation of a PCC system one must first review the factors that affect protein binding to resins. Protein adsorption to a chromatographic resin is dependent on the binding conditions, size and biochemical properties of the target protein, the resin bead morphology (bead size, pore structure and phase ratio), and the functional ligand properties (type and density). Collectively, these factors influence the thermodynamics and mass transfer kinetics of protein-resin interactions [23–31]. The thermodynamics govern the equilibrium binding properties, such as static binding capacity and the binding isotherm, while mass transfer kinetics govern the rate of protein adsorption onto the resin. Specifically, pore diffusion processes are present in most preparative scale chromatographic resins, which are porous beads having the majority of their surface area available within the pores [23]. The quantitative aspect of mass transfer kinetics in protein binding is demonstrated by the shape and nature of the protein breakthrough curve. Protein breakthrough from a column occurs before all the protein-accessible binding sites are utilized on the resin due to mass transfer resistance. Since columns are operated at a finite flow rate with a finite protein residence time, only a portion of the static binding capacity can be utilized. This capacity is the dynamic binding capacity at a particular residence time. Chromatographers have traditionally designed batch processes with relatively long target protein residence times to increase resin capacity utilization. However, this approach also results in low throughput and subsequent oversizing of the chromatographic column to achieve the desired target throughput. In contrast to batch chromatography, target protein breakthrough is loaded onto a second column in PCC until the first column is almost fully saturated. This leads to full capacity utilization irrespective of most commonly used residence times. This can also be achieved in batch mode in convective systems (perfusion media with large pores and membrane chro-

matography) at short residence times where the mass transfer effects are negligible.

The goal of this paper is to propose and apply a mathematical framework for designing PCC for a given protein-capture resin system. The scheduling required to perform continuous capture of a target protein using PCC is governed by the target protein breakthrough curve. For a given protein-capture resin system, the scheduling parameters can be determined from a mathematical transformation of the breakthrough curve. The PCC operation design range is also subject to additional constraints in order to achieve optimal operation. The optimal PCC operation can be defined using one of many potential objective functions, including maximization of resin capacity utilization, minimum recovery loss during wash, resin lifetime usage, or minimization of residence time. A formal approach for determining design constraints for optimal operation is also proposed and further verified with experimental results.

2 Materials and methods

2.1 Cell culture

Chinese Hamster ovary (CHO) cell lines producing mAb, Enzyme 1, or Enzyme 2 were grown in 12-L perfusion bioreactors (Broadley-James Corp., Irvine, CA, USA) in a proprietary chemically defined media. The cells were inoculated at 5×10^5 cells/mL and maintained at 5×10^7 – 6×10^7 cells/mL using a cell bleed apparatus. The cell bleed was used to maintain a constant cell density in the reactor to achieve near to steady-state conditions. The cell density was monitored offline by Vi-CELL[®] (Beckman Coulter, Brea, CA, USA) and online using Futura capacitance probes (Aber Instruments, Grand Island, NY, USA). An ATFTM system (Refine Technologies, Pine Brook, NJ, USA) containing a polyethersulfone 0.2- μ m tangential flow membrane was used to retain the cells in the reactor. The clarified harvest (permeate) was either collected into a 2-L harvest bag (Hyclone, Logan, UT, USA) or directly pumped to the PCC system. Additional details of the cell culture process and set-up were described previously [22].

2.2 Downstream

2.2.1 PCC operation

The 4-column PCC system used in this study was a custom modified ÄKTATM system (GE Healthcare). The system was equipped with five UV monitors (UV-900), three pumps (P-900), multiple valves (PV-908, SV-903), and a pH and conductivity meter (pH/C-900) and was powered by a UnicornTM-based custom control strategy (GE Healthcare). MabSelect SuReTM, iminodiacetate and hydrophobic interaction chromatography (HIC) media (all from GE Healthcare) were packed into three or four 1.6 cm $\times$ 6 cm,

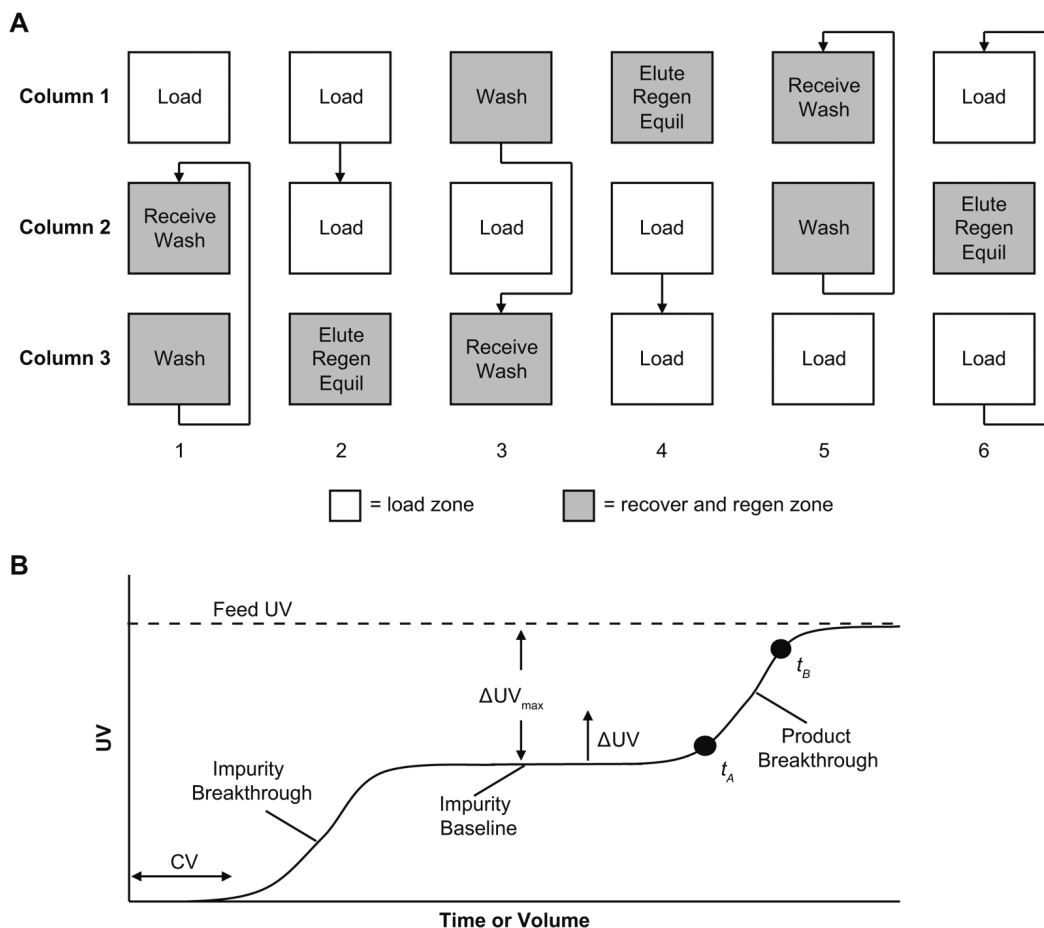


Figure 1. (A) Schematic diagram of the three-column PCC cycle. At the beginning of a cycle, the feed is loaded onto column 1 and the flow-through goes to waste (step 1). When the product starts to breakthrough, the flow-through from column 1 is directed to column 2 to capture the unbound product (step 2). Once column 1 is fully loaded, the feed is directly loaded onto column 2, while column 1 is washed onto column 3, eluted, regenerated (regen) and re-equilibrated (Equil) for the next cycle (steps 3 and 4). Column 2 is now subjected to steps 3–6, which are identical to steps 1–4 for column 1. Finally, the same steps occur on column 3. Note that the cycle restarts with column 1 in steps 1–4. (B) Column switching based on difference in feed and column outlet UV. t_A corresponds to the time when the ΔUV has reached a pre-determined threshold and the flow-through from column 1 is directed onto column 2. t_B denotes the time when the column has been saturated.

1.6 cm × 10 cm, or 0.66 cm × 6 cm columns according to the manufacturer recommendations and then connected to the PCC. The columns were qualified to meet internal HETP (height equivalent to a theoretical plate) and asymmetry specifications.

A chromatographic operation consists of equilibration, load, wash, elution and regeneration steps. In PCC, multiple columns are used to perform the same steps in parallel such that continuous feed is achieved and all other process steps are discrete in time. For ease of illustration, a three-column PCC operation is described in Fig. 1A. The continuous PCC operation can be broken down into subunits. For PCC, a cycle is defined when each one of the columns has completed equilibration, load, wash, elution and regeneration steps in sequence. To proceed from one step to another, an automated column switching algorithm is employed (as shown in

Fig. 1B). There are essentially two decision points required for every column cycle – one at the start of protein breakthrough and the other at column saturation. These switch points in PCC are determined by a novel UV-based control strategy developed by GE Healthcare that is based on a difference between the feed and column outlet UV, ΔUV . In principle, column switching can be determined by any PAT (process analytical technology) tool capable of in-line measurement of product concentration.

2.2.2 Breakthrough curves

A determination of the target protein breakthrough curve on the capture resin is required for PCC design. To obtain the breakthrough curves, frontal loading experiments [32] were performed for the three target proteins at two different residence times on 1–6-mL columns. The protein concentration was measured as described in Section 2.2.3.

2.2.3 Analytical methods

In-house assays were used for the quantitation of target protein in the column loads, flow-through, wash and eluate. mAb concentration was measured by HPLC using a Protein A column (Applied Biosystems, Carlsbad, CA, USA). For Enzyme 1 and 2, the enzymatic activity was determined by measuring the hydrolysis rate of an appropriate synthetic substrate linked to *p*-nitrophenol (pNP) (Sigma Aldrich, St. Louis, MO, USA). Samples (25 μ L) were incubated with 225 μ L 40 μ M substrate for 15 min at 37°C. The reactions were quenched with 250 μ L 0.3 M glycine, pH 10.5 and the absorbance was then measured at 400 nm. One unit of activity was defined as the amount of enzyme required to hydrolyze 1 μ mol substrate to pNP per minute.

2.3 Theory

This section outlines the design and operation of a continuous PCC capture step that is integrated with continuous perfusion cell culture. The conditions for optimal PCC operation for a given protein and capture resin are discussed with respect to resin capacity utilization, protein recovery and resin lifetime usage. Finally, the impact of process variability on PCC process design is briefly discussed using a design sensitivity analysis.

2.3.1 Feed continuity constraint

To allow for integration of continuous capture with a perfusion bioreactor, the discrete chromatography steps within a PCC operation must be scheduled such that a continuity of feed (column load) is maintained. PCC operation can be divided into two zones: (i) protein loading (zone – “load”) and (ii) all other steps (recovery, regeneration, zone – “recover-regen”). Multiple columns can be part of a zone and individual columns move in and out of a particular zone as a function of time (Fig. 1A). The feed continuity constraint requires that,

$$\tau_R' \leq \tau_L \quad (1)$$

where, τ_L is the average loading time of a column and τ_R' is the *recover-regen* (*R-R*) zone time for one column. For a three-column PCC, these times can be calculated using the following equations:

$$\tau_L = \frac{\kappa_S R_t}{C_f} \quad (2)$$

$$\tau_R' = t_w + t_e + t_r + t_{eq} \quad (3)$$

where κ_S is the static binding capacity of resin under the capture conditions, C_f is the target concentration in the feed and R_t is the target residence time. t_w , t_e , t_r and t_{eq} are wash, elution, regeneration and equilibration step times, respectively. These step times can further be divided into individual steps. For example, the column equi-

libration itself can consist of multiple steps requiring two or more buffers. Also, Eq. (2) assumes that binding capacity in PCC approaches the static binding capacity. If that is not the case, κ_S should be replaced by the actual binding capacity in PCC.

For an *N*-column PCC operation, the *R-R* time can be split across *N*-2 columns and is calculated as follows (2 columns are utilized for loading and washing):

$$\tau_R = \frac{t_w + t_e + t_r + t_{eq}}{N - 2} \quad (4)$$

For a given target protein-resin system, the feed continuity constraint can be satisfied by multiple PCC designs. To highlight the flexibility in designing the PCC operation within the feed continuity constraint, Eqs. (1–4) can be combined to yield the following equivalent relationships:

$$R_t > \frac{\tau_R C_f}{(N - 2)\kappa_S} \quad (5)$$

$$N > 2 + \frac{\tau_R C_f}{R_t \kappa_S} \quad (6)$$

The key design parameter that emerges from this analysis is the ratio of the static binding capacity to the feed concentration (κ_S/C_f). For the majority of protein-capture resin systems, this ratio is expected to fall within 5–100. For example, mAb processes typically use Protein A resins that have static binding capacities in the range of 45–60 mg/mL resin [24] for feed concentrations of >0.5 mg/mL, which results in a range for the ratio of 5–100. For a given protein, the κ_S/C_f ratio is known and constant once a capture resin is selected. Figure 2A illustrates the minimum residence time needed as a function of this ratio, thus showing the flexibility to select shorter residence times and, subsequently, smaller columns in PCC. Further, Fig. 2B shows the relationship between the number of columns, residence time and κ_S/C_f for integrated and continuous PCC operation. As this ratio increases, shorter residence times (or smaller columns according to Eq. 5) and fewer columns are needed to maintain feed continuity. For a given capture process, Eq. (6) also predicts that the number of columns increases linearly with feed concentration, a result previously reported for other types of multicolumn chromatography systems [13].

In addition, the feed continuity constraint defines both the cycle time and the size of each capture column in an *N*-column PCC system. The cycle time is the duration of one column operation and is given by τ_L according to Eq. (2). The size of each capture column (V_c) is defined by mass balance as:

$$V_c = D R_t \quad (7)$$

where *D* is the perfusion volumetric flow rate and R_t is the target protein residence time.

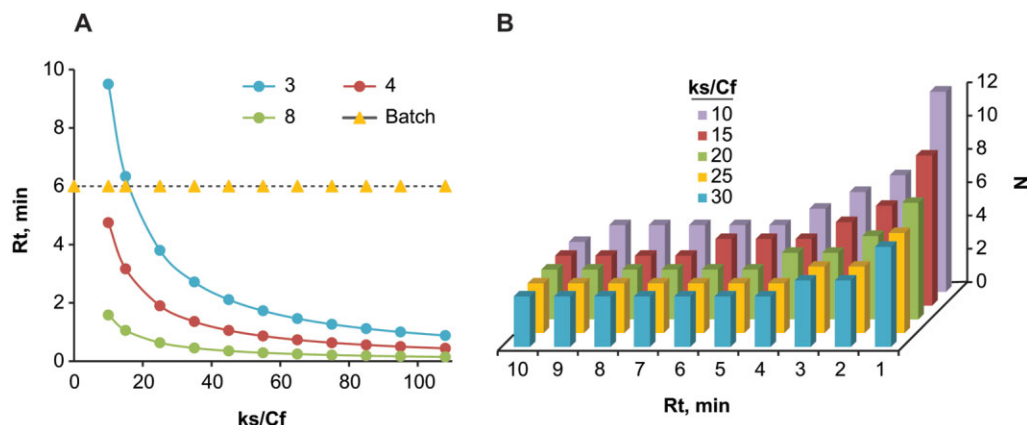


Figure 2. (A) Minimum residence time as a function of κ_s/C_f for an N-column PCC operation according to Eq. (5). Predicted data for $N = 3, 4$ and 8 are shown. The dotted line indicates a typical residence time used for a batch process. (B) Minimum number of columns required for a range of κ_s/C_f according to Eq. (6). For both figures, the value of τ_R was set as 95 min, assuming 15 min for wash, 30 min for elution, 30 min for regeneration and 20 min for re-equilibration.

2.3.2 Process optimization

2.3.2.1 Resin capacity

The binding capacity in PCC can be maximized such that it is close to the static binding capacity of the resin by loading the target protein breakthrough onto a second column (see schematic Fig. 3A). In an optimal PCC operation, maximum resin utilization, without any loss of protein on the second column, can be achieved if the dynamic binding capacity (κ_D) is greater than the breakthrough capacity from the first column (κ_2), which can be written as follows:

$$\text{Maximize, } U_{PCC} \text{ such that, } \kappa_2 = \int_0^{x_f} C_O(x) dx < \kappa_D \quad (8)$$

Where U_{PCC} is the resin capacity utilization in PCC and $U_{PCC} = \frac{\kappa_{PCC}}{\kappa_S} = \frac{\kappa_D + \kappa_1}{\kappa_S}$, κ_{PCC} is the resin binding capacity in PCC ($\kappa_D + \kappa_1$), and x_f is the total loading of a single column in PCC ($\kappa_D + \kappa_1 + \kappa_2$).

In order to calculate the parameters in Eq. (8), numerical integrations can be performed on an experimentally determined breakthrough curve. Alternatively, the protein breakthrough curve can be predicted and modeled using reasonable approximations for underlying thermodynamic and kinetic effects influencing the binding phenomena [24–26]. The goal of this paper is to illustrate PCC design using an experimental breakthrough curve. Therefore, an empirical model was used to fit experimentally determined breakthrough using the following equation:

$$C_O = \frac{C}{C_f} = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x - \kappa_S}{\sigma} \right) \right) \quad (9)$$

where C is the protein concentration at the column outlet and C_o is the normalized concentration at the column outlet, erf is error function, x is protein load (mg/mL resin) and

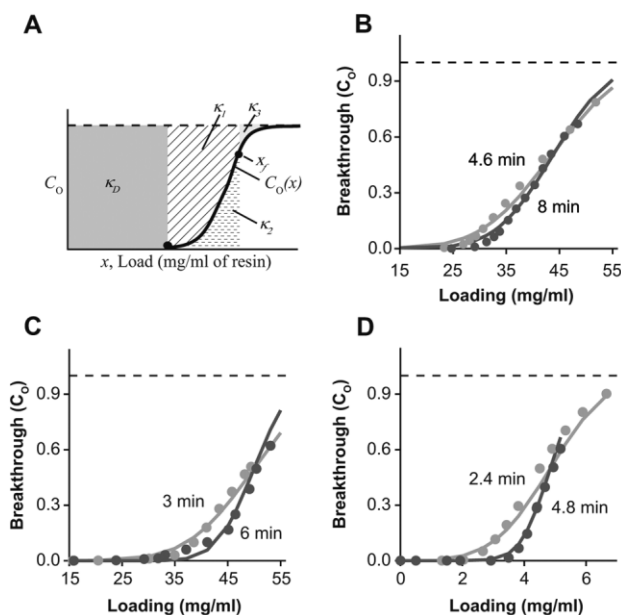


Figure 3. (A) Schematic of a breakthrough curve and important parameters for PCC design. The X axis represents target protein loading in mg/mL resin and Y axis represents the normalized outlet feed concentration and is unitless. (B) The breakthrough curve for Enzyme 1 on a pseudo-affinity capture resin, the points are experimental data and the lines are least square fits to Eq. (9). The data are shown at two different residence times: 8 min (dark gray, used in batch process) and 4.6 min (gray). The fitting parameters are given in Table 1. (C) The breakthrough curve for the mAb on affinity capture resin, points are experimental data and lines are least square fits to Eq. (9). The data are shown at two different residence times: 6 min (dark gray) and 3 min (gray). The feed concentration is about 0.4 mg/mL. (D) The breakthrough curve for Enzyme 2 on conventional HIC resin, points are experimental data and lines are least square fits to Eq. (9). The data are shown at two different residence times: 4.8 min (dark gray, used in batch process) and 2.4 min (gray). The error bars in measurement of breakthrough protein concentration are comparable to pointwidth as shown.

σ is an empirical constant that fits to the width of the sigmoidal breakthrough curve.

The value of resin loading, x , at any breakthrough concentration, C_o , can be approximated using the inverse error function (according to Eq. 9). By defining κ_D as 3% of protein breakthrough, κ_D can be approximated as $\kappa_s - 4/3\sigma$. Similarly, defining column saturation at 99% protein breakthrough, x_f can be approximated as $\kappa_s + 5/3\sigma$, which leads to κ_2 being equal to $5/3\sigma$. Using Eq. (8) and substituting the values for κ_D and κ_2 , we obtain the following equation for optimal PCC operation with resin capacity utilization at about 100%:

$$\sigma < \frac{\kappa_s}{3} \quad (10)$$

Eq. (10) is derived for the condition where the PCC binding capacity is approximately equal to the static binding capacity of the resin, and there is no loss of target protein in the first column flow-through because the target is captured by a second column. κ_s is a constant for a given protein-resin system under the capture conditions, and σ is an empirical constant that depends on residence time. An increase in residence time decreases σ , such that the condition in Eq. (10) becomes easier to satisfy. In cases where κ_s is so low that sufficient residence time cannot be achieved to meet this condition, a sub-optimal design has to be selected in which either a lower protein recovery or lower capacity utilization is achieved. The Enzyme 1 case study falls into this category and is discussed in greater detail in the Section 3.

2.3.2.2 Recovery and wash step

In batch chromatography, loading is limited to the point of breakthrough (underloading) such that all the target protein in the feed stream binds to the column. In contrast, column loading in PCC approaches resin saturation (overloading) and therefore, at the end of loading, the last column volume of the feed has approximately the same target protein concentration ($\sim C_f$) as in the feed. One or more wash steps are typically performed after loading and before elution to remove unbound or loosely bound impurities. In PCC, by having a second column in line for the wash step, the last column volume as well as the target protein present in the wash can be captured on to a second column. If the wash is not captured, the recovery loss is expected to be about C_f/κ_s . Capturing of wash on a second column creates additional scheduling constraints for PCC design. First, it requires that the PCC system consists of a minimum of three columns – one present in the *load* zone for continuous loading and two additional in the *R-R* zone to capture wash. This scheduling constraint for a three-column PCC can be expressed as:

$$t_w < \frac{(\kappa_d - \kappa_2)R_t}{C_f} \quad (11)$$

The approximations for κ_2 and κ_d , as estimated earlier, can be substituted in Eq. (11) to yield,

$$t_w < \frac{(\kappa_s - 3\sigma)R_t}{C_f} \quad (12)$$

The constraint in Eq. (12) results from the fact that only one column is available in the *load* zone during the wash step. In order to prevent loss of target protein, the column in the *load* zone should not reach its dynamic binding capacity (κ_D) prior to another column becoming available to capture breakthrough. This constraint is more restrictive than the constraint described in Eqs. (9) and (10). For an N -column PCC system (where $N > 3$), more columns are available to receive the wash and this constraint may be relaxed. In addition, the recovery loss due to wash can be higher than C_f/κ_s depending upon the shape of absorption isotherm under wash conditions.

2.3.2.3 Resin lifetime

An integrated PCC capture operation is expected to run for an extended period of time, which depends upon the duration of the perfusion bioreactor campaign. In the following, a campaign is defined as the entire duration of integrated PCC capture operation. The number of cycles for a single column during one PCC campaign can be expressed as:

$$\eta = \frac{M_c}{NV_c \kappa_s} \quad (13)$$

where η is the number of column cycles, M_c is the total mass of protein to be produced in one campaign, N is the number of columns in PCC design, and V_c is the volume of one column in PCC design.

For biotherapeutic manufacturing, a set number of column cycles (η_m) are performed prior to discarding the resin and packing a column with virgin resin. The maximum cycle number is based on important considerations, such as process-dependent resin deterioration related to lower target protein recovery or purity. The ability to select PCC parameters (R_t and N) such that $\eta \sim \eta_m$ allows for the maximization of resin use in one campaign, which is another advantage of PCC.

2.3.3 Design sensitivity analysis

The variability in feed concentration and column packing within the individual columns in the multicolumn system can potentially affect the PCC operation. If a high binding capacity resin is chosen and operated at a residence time that is longer than the minimum residence time needed (e.g. for $\kappa_s/C_f = 30$, $R_t > 4$ min; Fig. 2B), the PCC process is inherently robust to such variability. This may not always be practical and, as discussed below, a PCC operation can be designed to account for expected variability in these parameters.

2.3.3.1 Feed concentration

When the perfusion cell culture is operated and controlled at a quasi-steady state, the feed concentration is expected to remain relatively constant for the entire capture duration. However, the feed concentration may vary due to fluctuations in the cell culture process. Changes in feed concentration may result in changes in κ_S/C_f which affect the PCC operation as described in Sections 2.3.1 and 2.3.2. To understand the impact of feed concentration variation on the PCC process, binding isotherms can be performed to measure the static binding capacity as a function of feed concentration. For a variety of protein-resin systems, the binding isotherm can be fitted to Langmuir or Langmuir-like isotherms [24, 33–35]. For the Langmuir isotherm, κ_S is a weak function of C_f above a threshold feed concentration, and the κ_S/C_f is inversely proportional to C_f . Therefore, the PCC process can be designed conservatively using the lowest expected ratio to account for such variability. Conversely, if C_f is below the aforementioned threshold, the concentration variability is in the range where κ_S is proportional to C_f (linear regime of Langmuir Isotherm). In such cases, minimal change in κ_S/C_f is expected with variability in the feed concentration and the PCC process is expected to be robust.

2.3.3.2 Column packing

In a multicolumn PCC system, potential variations in column packing within the individual columns (i.e. compression factor and column height) can result in variable binding capacity and/or residence time on each column. These variations affect the continuous chromatography operation by breaking column periodicity, and have particular impact on continuous operation in systems where the column switching is based on time. In addition, this can also result in loss of target protein recovery if the residence or binding capacity does not meet the optimality criterion described in Section 2.3.2. In PCC, because of inline detection for column switching, minor variations in the period of each column are acceptable as long as the feed continuity and desired optimal constraints are met. Major variations in column packing can be accounted for using the lowest expected binding capacity and residence time during PCC design. Furthermore, column-to-column differences can be minimized by instituting a ro-

bust column packing procedure that leads to a reproducible and stable packed bed.

3 Results and discussion

Widespread implementation of an integrated and continuous capture process requires that it can be used for a wide range of proteins, chromatographic resins and process conditions. The case studies presented below cover a range of potential capture process characteristics and can be categorized in two main classes of proteins: (i) relatively unstable molecules captured by conventional resins (enzymes) and (ii) a relatively stable molecule captured with an affinity resin (mAb). The initial work focused on the development of a three-column PCC methodology for capturing Enzyme 1. The feasibility of integrating the PCC with the perfusion bioreactor and running continuously for an extended period of time was then demonstrated with the mAb and Enzyme 2.

3.1 Method development

The custom GE Healthcare PCC system uses a novel UV-based control strategy that allows for automated column switching based on breakthrough loading. A relatively simple protein-capture system was chosen for initial method development – capture of Enzyme 1 on a pseudo-affinity resin. The Enzyme 1 load was relatively high in concentration (~1 mg/mL) and, therefore, changes in column outlet concentration were easily detected by UV. In addition, Enzyme 1 was the primary component in the UV feed signal, comprising >85% of the absorbance. This resulted in minimal UV signal interference from impurities and, correspondingly, from the impurity baseline.

For method development, the selection of residence time, capacity utilization and number of columns in PCC operation is needed. These parameters are dependent on the constraints and can be established using an experimentally determined breakthrough curve (Section 2.3). The Enzyme 1 breakthrough curves were determined at 8.0 min (used in batch process) and 4.6 min residence times under the capture conditions (Fig. 3B). In this range of residence time, the shape of the breakthrough curve

Table 1. Fitting parameters for breakthrough curve for various protein-resin systems. The empirical parameters κ_S and σ have error bars of $\pm 5\%$ of their regressed value

Protein	Resin	C_f (mg/mL)	R_t (min)	κ_S (mg/mL)	σ (mg/mL)	κ_D (mg/mL)
Enzyme 1	Pseudo-affinity	1.0	4.6	43.6	15.4	23.1
Enzyme 1	Pseudo-affinity	1.0	8.0	43.6	12.7	26.7
mAb	Affinity	0.4	3.0	50.0	14.1	31.2
mAb	Affinity	0.4	6.0	50.0	8.0	39.3
Enzyme 2	HIC	0.3	2.4	4.8	2.1	2.0
Enzyme 2	HIC	0.3	4.8	4.8	1.1	3.3

Table 2. Conditions for maximum capacity utilization (~100%) in PCC operation for Enzyme 1^{a)}

Residence time (R_t , min)	Feed continuity		Maximum resin utilization		Capturing wash	
	τ_R , min	τ_L , min	σ , mg/mL	$\kappa_s/3$, mg/mL	t_w , min	$(\kappa_s - 3\sigma)R_t/C_f$, min
4.6	90	201	15.4	14.5	11	-12
8.0	90	349	12.7	14.5	11	44

a) The values were calculated using Eqs. 2, 10 and 12 for feed continuity, maximizing resin utilization and capturing target in the wash, respectively.

was approximately sigmoidal and fitted well to Eq. (9). As expected, with increasing residence time, the dynamic binding capacity increased and the transition from κ_D to κ_S was sharper. This behavior was also reflected quantitatively in parameter fitting (Table 1) where the value of σ decreased with an increase in residence time. These fits were instructive for quantitative comparison among various protein-capture resin systems and also enabled quick calculation of the integrals detailed in Section 2.3.

The σ and κ_S (Table 2) determined from breakthrough curve were used to calculate the conditions for the feed continuity constraint (Eq. 1) while maximizing both resin capacity utilization and target recovery by capturing the wash (Eqs. 8 and 11). For Enzyme 1, the calculated κ_S/C_f was 44 and, therefore, the feed continuity constraint was met at both the residence times. Even though the binding thermodynamics were favorable, resulting in high Enzyme 1 binding capacity (44 mg/mL), the slow binding dynamics for this system resulted in a high value of σ for residence times of ≤ 4.6 min (Table 2). As a result, the two conditions (Eqs. 10 and 12) for maximum resin utilization and capture of wash were not met at the lower residence time of 4.6 min. If 100% resin utilization in PCC is desired, the longer residence time of 8 min may be needed for this and similar protein-capture resin systems.

According to Eq. (7), the benefits of operating at shorter residence times are significant in terms of reduced column volume and buffer consumption. To realize these benefits for Enzyme 1, however, at least one of the optimality constraints – maximum resin utilization or capture of the wash – must be violated, which leads to either sub-optimal resin utilization or product loss. The breakthrough curve fitting parameters, σ and κ_S , were used to determine the normalized breakthrough concen-

tration, C_O , for 95% capacity utilization according to Eq. (9). C_O was 0.69 for 4.6 min residence time and 0.64 for 8 min. In contrast, the maximum capacity utilization at 4.6 min residence time is expected to be less than 50% (κ_D/κ_S) for a fully optimized batch process. The small difference in C_O at the significantly different residence times indicated that most of the target protein in the load was breaking through near the end of the first column loading. As a result, a significant capacity of the second column (κ_2) was consumed for only a marginal gain in capacity utilization on the first column (κ_1). More generally, this interplay between capacity utilization and throughput (residence time) can be exploited in a much broader range for a PCC capture process when compared to the batch process, resulting in both high resin capacity utilization and high process throughput, even in sub-optimal PCC designs.

The preceding analysis was used to design and execute the three-column PCC capture of Enzyme 1. Specifically, the method used a residence time of 4.6 min and a ΔUV cut-off for column saturation at 70%. Based on this, the theoretical model predicted feed continuity, 95% resin capacity utilization and recovery comparable to batch operation (by capturing wash, Table 3). Figures 4A and B show the $A_{280\text{ nm}}$ chromatographic profile for the PCC capture of Enzyme 1. The recovery (80%) and purity were consistent across all three columns in the PCC (data not shown). By loading protein breakthrough to a second column, about 40 mg/mL of resin binding capacity was utilized. These data showed the successful application of theoretical continuous processing concepts, including feed continuity, optimal resin utilization (95%) and optimal recovery, during the design of a PCC methodology for the capture of Enzyme 1 on a pseudo-affinity resin.

Table 3. Conditions for 95% capacity utilization for PCC operation for Enzyme 1^{a)}

Residence time, (R_t , min)	Feed continuity		Optimal resin utilization		Capturing wash	
	τ_R , min	τ_L , min	κ_2 , mg/mL	κ_D , mg/mL	t_w , min	$(\kappa_D - \kappa_2)R_t/C_f$, min
4.6	90	191	7.6	23.1	11	71
8.0	90	331	5.5	26.7	11	169

a) The values were calculated using equations 1, 8 and 11 for feed continuity, optimal resin utilization and capturing target in the wash, respectively. The κ_2 was calculated using numerical integration of fitted model.

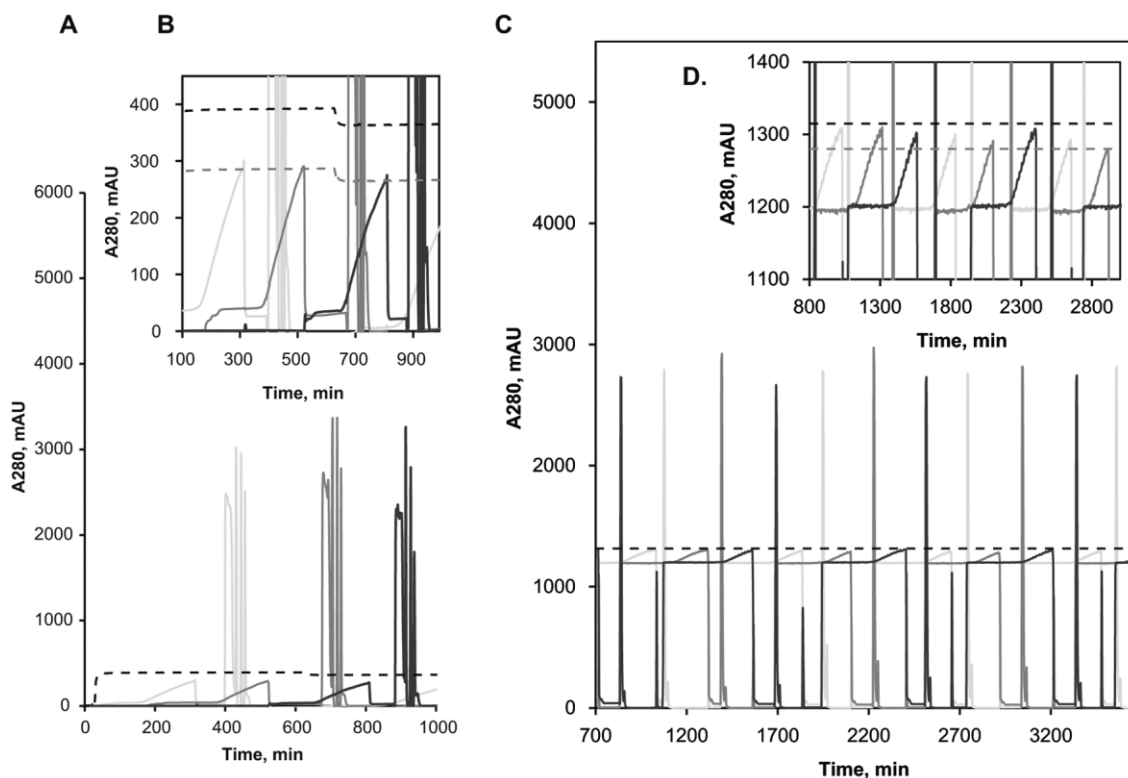


Figure 4. (A, B) Chromatographic profiles for PCC capture of Enzyme 1 using a three-column PCC methodology, including feed UV (black dashed), column 1 UV (light gray solid), column 2 UV (gray solid) and column 3 UV (black solid). (B) Magnification of the breakthrough to illustrate automated column switching based on UV triggers. At 70% ΔUV (gray dashed line), the first column was switched from feed to wash and the Enzyme 1 in the wash was captured by the third column. The small dip in feed UV at about 600 min corresponds to changing to a lower target protein concentration feed material. (C, D) Chromatographic profiles showing $A_{280\text{ nm}}$ trace for continuous capture of mAb on mAbSure™ using a three-column PCC, including the feed UV (dashed line) and the outlet UV for the three columns (solid lines). A snapshot for 2 days is shown. (D) The same data on a different scale to highlight the variability in switch times for column saturation. The gray dotted line represents 70% ΔUV . The automated switch corresponding to column saturation point was variable and occurred mostly above the target 70% mark due to a known issue in the current version of the control strategy. The error bars in UV measurement are comparable to linewidth as shown.

3.2 Case studies

The proposed methodology was also used to investigate the feasibility of performing integrated, automated and continuous capture for an extended period of time. The goal was to monitor process performance over time for a relatively stable protein (mAb) and a relatively unstable protein (Enzyme 2), and to gain an understanding of the robustness of the PCC system hardware and software. In contrast to Enzyme 1, both the feed concentration (0.3–0.5 mg/mL) and the ratio of target protein to impurities (~10%) was considerably lower for the mAb and Enzyme 2. Accurate determination of ΔUV is challenging when both the protein concentration as well as the ratio of target protein to impurities are low. This challenge was mitigated by the development of a robust method to calibrate the UV monitors. Each monitor was calibrated with acetone and the path lengths of the UV monitors were manually adjusted so that they would all have the same output.

Figures 3C and 3D show breakthrough curves for mAb on a Protein A resin and Enzyme 2 on a HIC resin at two residence times, respectively. The static binding capacity (Table 1) on the Protein A resin was about 50 mg/mL. Under the capture conditions for Enzyme 2, the binding capacity on the HIC resin was low: 4.8 mg/mL resin. The feasibility of continuous capture for both of these proteins is discussed in separate subsections below.

3.2.1 mAb

A κ_S/C_f of 100 was calculated for the mAb (Table 1). This indicated that a three-column PCC would be sufficient to satisfy the feed continuity constraint. In fact, using σ and κ_S from the breakthrough curve, all conditions were met for 100% PCC capacity utilization (Eqs. 1, 10 and 12) at both 3 and 6 min residence time. This indicated that three columns and a residence time <3 min were sufficient for continuous capture of mAb. Based on this, a residence time of 2.5 min was selected for this mAb. To test the long-term performance of the PCC system, bioreactor harvest

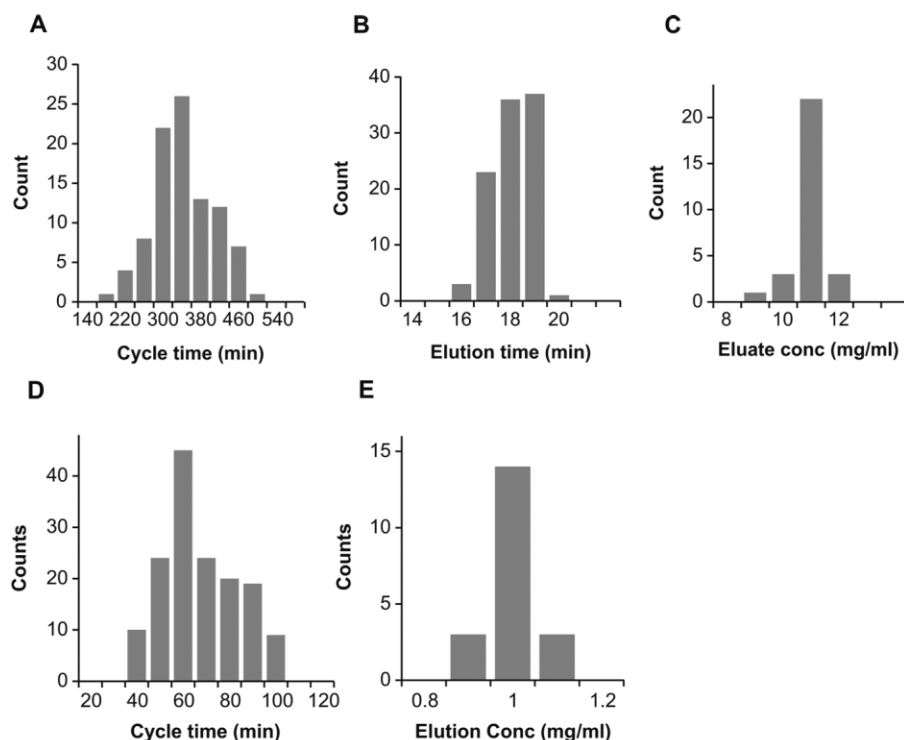


Figure 5. (A) Distribution of individual column cycle time over 30 days of continuous capture of mAb on affinity capture step using a three-column PCC. The cycle time was calculated from the chromatographic profile as the time between two consecutive elutions on a given column. The first and second moments of this distribution were 323 and 64 min, respectively. (B) Distribution of elution time (or volume) over 30 days of continuous operation. The first and second moments of this distribution were 17.6 and 0.9 min, respectively. (C) Distribution of capture eluate protein concentration by $A_{280\text{ nm}}$ over 30 days. The first and second moments of this distribution were 10.4 and 0.5 mg/mL, respectively. Three to five elutions were pooled approximately daily before concentration measurement. (D) Distribution of individual column cycle time (period) over 9 days of continuous capture of Enzyme 2 on an HIC column using a four-column PCC operation. The first and second moments of this distribution were 62 and 16 min, respectively. (E) Distribution of capture eluate protein concentration over 9 days. The first and second moments of this distribution were 0.94 and 0.05 mg/mL, respectively. The protein concentration was measured in column eluates from PCC cycles 1, 2, 3, 10 and 40.

was continuously captured for 30 days, which corresponded to 114 column operations. A total of 228 automated column switches occurred over a period of 30 days and the reproducibility between cycles was demonstrated in the chromatogram snapshot shown in Figs. 4C and D. Using this methodology, the resin capacity in PCC was comparable to the static binding capacity (~49 mg/mL resin).

To further investigate the consistency of column switching and the performance of individual columns in PCC, the chromatographic profiles were analyzed to calculate cycle time, elution time (or volume) and eluate protein concentration over the entire operation. The calculated distribution of cycle time is shown in Fig. 5A. The calculated average cycle time was 323 ± 64 min, which was in close agreement with the predicted cycle time of 313 min using Eq. (2). The variability in cycle time can further be minimized by upgrading to an improved version of the UV-based control strategy. The analysis of elution volume and eluate concentration (Figs. 5B and C) revealed a tight distribution for both parameters, indicating uniform target protein loading across columns cycles. The

critical quality attributes of the mAb, including potency, purity, host cell protein, residual Protein A and aggregation levels were comparable over the entire duration of PCC operation [22]. In summary, these results demonstrate the feasibility of integrating continuous mAb capture from a perfusion bioreactor using PCC.

3.2.2 Enzyme 2

In contrast to the mAb case study, the feed continuity constraint for the three-column PCC was not met at a residence time of 2.4 min, and was only marginally met at a residence time of 4.8 min due to the low κ_S/C_f (Eq. 1; with $\tau_R = 68$ min). This constraint can be satisfied by either increasing the residence time beyond 4.8 min or adding a fourth column to the PCC design. In this study, the feed continuity constraint was met by adding a fourth column without changing the residence time. By the addition of fourth column, the R - R zone was split onto two columns, reducing the τ_R by a factor of two. Thus, the four-column PCC methodology was developed using a 4.8 min residence time and other optimal constraints were the same as used for Enzyme 1. According to Eq. (9), the column

Table 4. Benefits of PCC capture over batch process

System	Resin capacity utilization (λ) ^{a)}	Buffer consumption ($1/\lambda$, %)	Column size reduction ^{b)} (ζ , fold)
Enzyme 1 pseudo-affinity	3.3	30%	35
mAb affinity	1.3	80%	26
Enzyme 2 HIC	2.0	50%	23

a) Unitless.

b) The size reduction factor was calculated in comparison with existing batch process. The smaller columns are cycled faster in PCC and the savings in resin and buffer consumption are thus scaled by λ (not by ζ).

saturation switching set point was determined to be 58% of the normalized outlet concentration, C_0 .

The four-column PCC was integrated with a perfusion bioreactor, and continuous capture was performed for 9 days, resulting in 166 column operations. During this operation, a total of 332 automated column switches occurred with an average cycle time of 62 ± 16 min (Fig. 5D). This experimentally determined cycle time was in good agreement with the predicted value of 63 min (Eq. 2). The analysis of eluate concentration (Fig. 5E) showed a tight distribution, indicating uniform target protein loading across columns cycles. As expected, the resin capacity in PCC was comparable to the static binding capacity (4 mg/mL resin). The recovery and all critical quality attributes were comparable across the four columns and throughout the entire period of harvest capture [22]. These results further demonstrate the feasibility of the continuous capture of a relatively unstable enzyme on a non-affinity (HIC) resin for an extended period of time. Because no time-based performance decline was observed in these case studies, it is possible to extend the integrated and continuous capture beyond the current study duration.

3.3 Advantages of PCC over batch chromatography

The integrated and continuous biomanufacturing platform offers several advantages over current biomanufacturing platforms from a facility and process design perspective [22]. Specifically, use of PCC for downstream processing allows process simplification by elimination of non-value-adding unit operations (harvest hold, microfiltration), reduction in column size, high resin capacity utilization and concomitant decreased specific buffer consumption. The column size reduction factor and gain in resin capacity utilization can be calculated according to:

$$\zeta = \frac{\tau_H C_f}{R_t \kappa_B} \quad (14)$$

$$\lambda = \frac{U_{PCC}}{U_B} \quad (15)$$

where, ζ , is the reduction factor (fold reduction) in column size for a continuous capture and τ_H is the harvest hold time for a batch chromatographic process, λ is the relative gain in resin capacity utilization in PCC with respect to

batch mode, U_B is the capacity utilization in batch mode $= \kappa_B / \kappa_S$, and κ_B is the binding capacity in batch mode which is generally $< \kappa_D$.

Table 4 shows the column size reduction factor, gain in resin capacity and corresponding reduction in buffer usage for the three case studies presented in this paper. The size of capture column was reduced by more than one order of magnitude (23–35). The resin capacity utilization was increased by a factor of 1.3–3.3, with concomitant decrease in buffer usage.

4 Concluding remarks

Integrated and continuous processing biomanufacturing has several advantages over the traditional batch or semi-batch platform typically used in production of biotherapeutics. In this work, we describe the principles and practice for PCC operation. A step-by-step approach for the design of a PCC operation was successfully applied at development scale to biotherapeutics with diverse properties, such as a high-volume more stable protein (mAb) and low-volume less stable proteins (Enzymes 1 and 2). The PCC process performance indicators, such as cycle time, elution volume, and eluate concentration, were consistent over an extended period of time.

Rapidly improving cell culture productivity has shifted much of the cost and throughput burden onto the downstream processes. The use of PCC can result in an automated highly productive capture step that uses small columns, fast cycling and results in greater throughput and a significantly reduced facility footprint. The PCC also reduces buffer and resin consumption, both of which are important cost factors in downstream processing. The reduction in column size is particularly relevant to mAb clinical manufacturing, where relatively large columns are packed with expensive affinity resins and often used for only a few cycles. The ability to use smaller columns and faster cycling in PCC would result in efficient use of the resin to the end of its life during a clinical campaign.

The true potential of continuous chromatography can only be realized by integration with a continuous upstream process. The main challenges for an integrated and continuous manufacturing process are the mechanical and microbial robustness of the systems. Specifically, the systems must operate continuously for months with-

out mechanical breakdown while maintaining a low bioburden environment. We believe that the solution to these challenges lies in a robust skid based on a closed and aseptic design. Once these challenges are met, the natural next step will be to extend the continuous operation boundary to further downstream steps, moving further towards achieving a fully continuous process for the production of biotherapeutics.

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Nomenclature

$C(x)$	[mg target protein/mL]	Column outlet concentration
C_f	[mg target protein/mL]	Feed concentration
C_0	[-]	Dimensionless concentration of target protein at the column outlet
D	[mL/min]	Bioreactor perfusion flow rate
erf	[-]	Error function
M_c	[mg target protein]	Total mass produced in one campaign
N	[-]	Number of columns in PCC system
R_t	[min]	Residence time during loading
t_w, t_e, t_r and t_{eq}	[min]	Wash, elution, regeneration and equilibration step times, respectively
U_B	[%]	Capacity utilization in batch mode
U_{PCC}	[%]	Capacity utilization in PCC
V_c	[mL resin]	Volume of individual PCC column
x	[mg target protein/mL resin]	Column loading
x_f	[mg target protein/mL resin]	Total loading of single PCC column
ζ	[-]	Column size reduction factor for continuous capture
η	[-]	Number of column cycles in PCC campaign
η_m	[-]	Maximum resin cycle number
κ_B	[mg target protein/mL resin]	Binding capacity in batch mode
κ_D	[mg target protein/mL resin]	Dynamic binding capacity
κ_S	[mg target protein/mL resin]	Static binding capacity
κ_1	[mg target protein/mL resin]	Additional binding capacity utilized due to breakthrough loading
κ_2	[mg target protein/mL resin]	Binding capacity used onto second column in breakthrough loading
κ_3	[mg target protein/mL resin]	Un-utilized binding capacity
λ	[-]	Gain in resin capacity utilization in PCC relative to batch
σ	[mg target protein/mL resin]	Empirical constant fit to width of sigmoidal breakthrough curve
τ_H	[min]	Harvest hold time in a batch chromatographic process
τ_L	[min]	Total load time for a single column
τ_R	[min]	Time required to wash, elute, regenerate and re-equilibrate a single column in 3-column PCC
τ_R	[min]	Time required to wash, elute, regenerate and re-equilibrate a single column in a generic, N-column PCC

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